ML MINI PROJECT

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SEC-B

Application of Machine Learning Methods to Identify and Categorize Radio Pulsar Signal Candidates

1. Problem Statement

Radio pulsars are rapidly rotating neutron stars that emit periodic radio signals. Detecting genuine pulsar signals from radio telescope observations is challenging because the data contains many non-pulsar signals and noise. Manually examining these candidates can be time-consuming.

The objective of this project is to apply machine learning techniques to classify radio signal candidates as either genuine pulsars or non-pulsars using statistical features extracted from their signals.

2. Dataset Description

This project uses the HTRU2 dataset, which contains **17,898 samples** and eight numerical input features for each sample. These features describe the statistical properties of the integrated pulse profile and the dispersion measure–signal-to-noise ratio (DM-SNR) curve.

The dataset contains two classes:

- **Non-pulsar (0):** 16,259 samples.
- **Pulsar (1):** 1,639 samples.

The dataset is imbalanced because non-pulsar samples significantly outnumber genuine pulsar samples. Therefore, evaluating the models using recall, precision and F1-score, in addition to accuracy, is important.

3. Methodology

The dataset was loaded and analyzed using Python. The input features were separated from the target labels, and the data was divided into training and testing sets using an 80:20 split. Stratified sampling was used to preserve the class proportions in both sets.

Feature standardization was applied for models sensitive to feature scales. Four model configurations were evaluated:

1. **Logistic Regression:** Used as a baseline classification model.
2. **Support Vector Machine (SVM):** Used to classify candidates by learning a decision boundary between the two classes.
3. **Random Forest:** Used as an ensemble model that combines predictions from multiple decision trees.
4. **Random Forest with SMOTE:** SMOTE was applied only to the training data to balance the minority and majority classes before training the Random Forest model.

The models were evaluated using accuracy, precision, recall and F1-score. Confusion matrices were also examined, and ROC curves were plotted to compare model performance.

4. Implementation and Results

The project was implemented using Python and libraries including Pandas, NumPy, Matplotlib and Scikit-learn, along with the imbalanced-learn library for SMOTE.

The following results were obtained on the held-out test dataset:

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	97.96%	94.43%	82.62%	88.13%
SVM	98.07%	94.81%	83.54%	88.82%
Random Forest	98.02%	91.86%	85.98%	88.82%
Random Forest + SMOTE	97.46%	83.95%	89.33%	86.56%

The SVM achieved the highest accuracy of 98.07%, while Random Forest with SMOTE achieved the highest recall of 89.33%.

SMOTE increased the model's ability to identify genuine pulsars, although precision and overall accuracy decreased. This demonstrates the trade-off between identifying more genuine pulsars and reducing false-positive predictions.

5. Conclusion

This project demonstrates that machine learning can effectively distinguish pulsar candidates from non-pulsar signals using statistical features. All four model configurations achieved test accuracy above 97%.

The results show that accuracy alone is not sufficient for evaluating a model on an imbalanced dataset. Random Forest with SMOTE achieved the highest recall, making it useful when missing a genuine pulsar candidate is considered particularly costly. SVM achieved the highest overall accuracy among the tested configurations.

The project provides a practical comparison of supervised classification techniques and demonstrates how class imbalance affects model performance. The current implementation performs binary classification; identifying specific pulsar types would require additional relevant data and further analysis.